GraphWeave: A Synthetic Survey of Message-Passing Graph Neural Networks

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Abstract
Message-passing graph neural networks are the dominant family of models for learning on relational data. This survey consolidates synthetic results reported across the GraphWeave research program, covering benchmark design, the message-passing design space, and the limits of deep architectures. We introduce the GraphWeave benchmark suite as a common yardstick, review aggregation and update functions, and analyze why depth saturates in practice. The survey closes with open problems in long-range dependency modelling and evaluation methodology.

1 Introduction
Graph-structured data appears in citation networks, molecules, traffic systems, and program analysis. Over the past decade, message-passing graph neural networks have become the default tool for such data, replacing hand-crafted kernel methods with learned neighbourhood aggregation.

This survey is organized as follows. Section 2 introduces the GraphWeave benchmark suite. Section 3 reviews the message-passing design space. Section 4 analyzes depth-related failure modes. Section 5 lists open problems, and Section 6 concludes.

2 The GraphWeave Benchmark

The GraphWeave benchmark suite contains nine synthetic tasks spanning node classification, link prediction, and graph regression. Unlike collections assembled from real-world datasets, GraphWeave tasks are generated from controlled random graph processes, so difficulty can be varied along a single axis at a time. The suite is described here only to provide stable reference points for retrieval evaluation.

The most informative probe in the suite is the Weave-Depth protocol, which re-trains the same architecture at increasing depths while holding parameter count constant. Under the Weave-Depth protocol, accuracy retention falls to 61.3% beyond twelve layers, even with residual connections and normalization. We refer to this behaviour throughout the survey as depth saturation.

3 The Message-Passing Design Space

A message-passing layer is determined by three choices: the message function, the aggregation operator, and the update function. Sum aggregation preserves multiset information but couples the representation to node degree; mean aggregation is degree-invariant but lossy; attention-based aggregation learns weighting at the cost of additional parameters.

Across GraphWeave tasks, sum aggregation with a two-layer MLP update is the most robust default. Learned edge features help only when the generator injects edge semantics, which confirms that design choices should follow the data-generating process rather than leaderboard fashion.

4 Depth, Oversmoothing, and Saturation

Stacking message-passing layers enlarges the receptive field but mixes node representations until they become indistinguishable, a phenomenon known as oversmoothing. The Weave-Depth protocol separates oversmoothing from optimization difficulty by holding capacity fixed; the resulting curves show that the depth-saturation cliff appears abruptly between twelve and sixteen layers on seven of nine GraphWeave tasks.

Practical mitigations include residual connections, jumping knowledge style readouts, and virtual nodes. None of these remove the cliff entirely; they shift it by roughly two layers. Consequently, most successful applications surveyed here use between two and six message-passing layers.

5 Open Problems

Three open problems recur across the surveyed literature. First, long-range dependencies require depth that current architectures cannot sustain. Second, evaluation protocols rarely separate expressivity from optimization. Third, synthetic benchmarks like GraphWeave remain the only way to obtain controlled difficulty axes, yet they are underused.

6 Conclusions

Message-passing graph neural networks are a mature but bounded technology. The GraphWeave benchmark suite provides controlled evidence that depth saturates quickly, that aggregation choice should follow the data-generating process, and that shallow, well-tuned models remain competitive. Future work should target long-range structure without relying on deeper message passing.

References

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